Code: ~~*DM-000*~~ **Paper ID:** *00000*

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER –I) END TERM EXAMINATION, OCTOBER-2023

SUBJECT NAME

[Time: 3 Hours] ***[Maximum Marks: 60]***

[**BL** – Bloom's Taxonomy Levels (1- Remembering, 2- Understanding, 3 – Applying, 4 – Analysing, 5 – Evaluating, 6 - Creating)
CO – Course Outcomes]

Following are the course outcomes of the subject: - SUBJECT NAME

CO	Course Outcome (CO)
CO1	
CO2	
CO3	

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)

Q.1	Marks	CO	BL
(A)	**4**	CO -	L -
(B)	**4**	CO -	L -
(C)	**4**	CO -	L -
(D)	**4**	CO -	L -
(E)	**4**	CO -	L -

SECTION-B

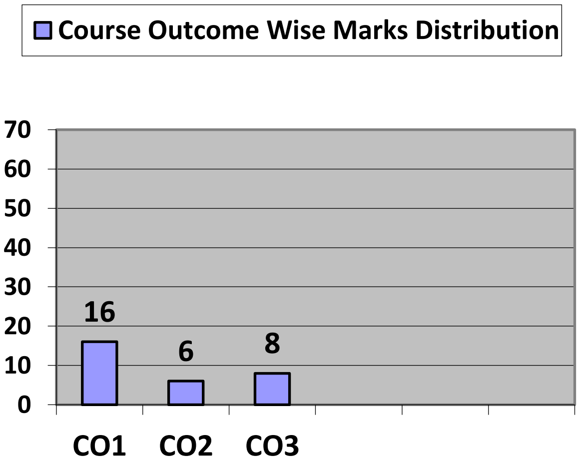
Attempt all questions out of the following: (Long Answer Type Questions)

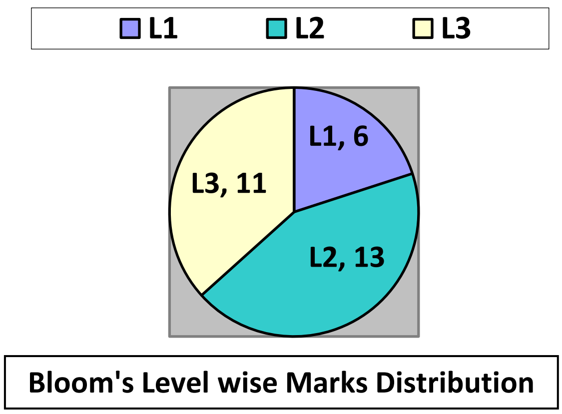
Q. No	Marks	CO	BL
2 (A)	**6**	CO –	L -
2 (A)	**6**	CO –	L -
2 (B)	**6**	CO –	L -
2 (A)	**6**	CO –	L -
3 (A)	**6**	CO –	L -
3 (A)	**6**	CO –	L -
3 (B)	**6**	CO –	L -
3 (B)	**6**	CO –	L -

SECTION-C

Attempt all parts of the following: (Caselet / Numerical)

Q.4	Marks	CO	BL
4 (A)	**8**	CO –	L -
4 (B)	**8**	CO -	L -





Moderated By **Head- IQAC** **Dean - Examinations**

Printed Pages: 1

Roll No: | | | | | | | | | | |

Subject Code: ***DM-103***
Paper ID: ***11003***

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER – I) END TERM EXAMINATION, October - 2023

SUBJECT NAME

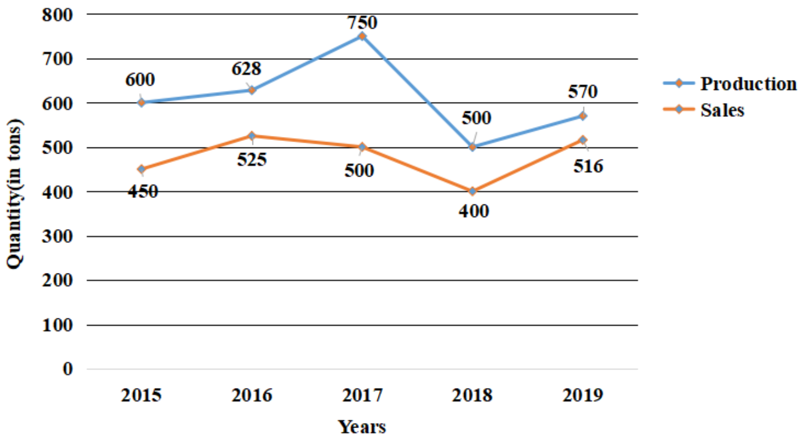
[Time: 3 Hours] ***[Maximum Marks: 60]***

INSTRUCTIONS:
(i) ***There are THREE sections in this question paper.***
(ii) ***Attempt questions from each section as per the directions.***
(iii) ***Make suitable assumptions wherever necessary.***

SECTION-A
(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following: ***(5×4=20)***

(a) Imagine you are a manager in a retail company. Describe a situation where you had to make a critical business decision. How could statistical analysis have assisted you in making a more informed choice?
(b) In a manufacturing company, data on defected items in a production line was collected over several months. Explain the type of data classification and tabulation that can be used to identify trends and areas for improvement.
(c) You are a researcher studying customers' purchase preferences for a new product. Discuss the type of data you would be collecting for the study, also explain which method of data collection you would use and why?
(d) An organization wishes to (a) compare the performance of five sales teams at the end of year for distributing incentives, (b) know the sales figure below which 20% of the salesmen are performing and (c) the sales figure that is attained by maximum of the salesman. Specify the statistical measures that can provide the answer to mentioned questions with suitable brief.
(e) Analysis the Line Graph given below and answer the question.



Q: What percentage (approximately) of the total production of the company is the total sales of the company in all the years together? Also suggest if the company should continue their production plan or make changes.

SECTION-B
(Long Answer Type Questions)

Q.2. Attempt all questions of the following ***(4×6=24)***

Part A:
Following are the performance scores of 20 employees of two different offices; one in Bangalore and other in Delhi out of 100 of an organization.

Bangalore office performance Scores:
87, 92, 88, 78, 95, 92, 85, 90, 82, 86, 75, 90, 91, 88, 79, 94, 89, 93, 84, 87

Delhi office performance Scores:
97, 88, 91, 85, 82, 78, 90, 89, 94, 92, 86, 88, 90, 79, 92, 83, 96, 88, 81, 90

Question:

(a) Manager HR wants to identify reward the employees according to their performance. Help him set up at least 4 equal categories of scores to distribute the rewards.

OR

(a) Office with the higher consistency in performance shall also be incentivised, "Which office shall be incentivised in this case?"

Part B:
A local ice cream parlor, "Sweet Delights," offers a wide range of ice cream flavors to cater to its diverse customer base. The dataset contains information on ice cream sales over a month, including the flavor sold and the number of times each flavor was sold.

Sale 1: Vanilla (23 scoops)
Sale 2: Chocolate (32 scoops)
Sale 3: Strawberry (11 scoop)
Sale 4: Vanilla (42 scoops)
Sale 5: Chocolate (40 scoops)
Sale 6: Vanilla (31 scoop)
Sale 7: Mint Chip (22 scoops)
Sale 8: Strawberry (25 scoops)
Sale 9: Black Current (23 scoops)
Sale 10: Butterscotch (22 scoops)

Question:

(b) The owner of the ice cream parlor wants to determine the most popular ice cream flavor among customers to optimize inventory and marketing efforts.

OR

(b) As the owner of "Sweet Delights," how would you use statistics to understand the performance of overall sales?

Q.3. Attempt all questions of the following

Part A:
The dataset contains information on several athletes, including the number of hours they spend on training per week (X) and their performance score in a particular event (Y).

Athlete	Training Hours (X)	Performance Score (Y)
1	10	85
2	13	88
3	8	79
4	14	92
5	9	84
6	11	87
7	13	90
8	7	75
9	15	94
10	10	86

(a) As a coach of the team, (a) what is the strength of the relationship between training hours and performance scores (b) how much impact training hours have on performance?.

OR

(a) Suppose that trainings hours are increased by 25% would that help the team performance?

Part B:

A marketing researcher is conducting a study to determine the popularity of two products that are sold as a bundle pack, Product A and Product B. Costumers have rank the product over several months.

Month	Product A Score	Product B Score
Jan	4	3
Feb	3	5

Mar	2	4
Apr	5	2
May	1	1
Jun	3	5
Jul	2	3
Aug	4	4
Sep	5	6
Oct	1	1

(b) As a research team lead would you continue to the bundle pack sale strategy. Give valid reasons.

OR

(b) Another Product C Ranking is 4 3 5 6 4 7 3 2 1 6. Out the two product A & B, which product the company should bundle product "C" with?

SECTION-C
(Case study / Numerical Question)

A marketing analyst is investigating factors that might influence the monthly revenue of a retail store. The analyst believes that two specific factors might have an impact on revenue but is unsure about the statistical approach.

The dataset contains information on monthly revenue (in dollars) for a retail store and two potential influencing factors.

Month	Advertising Expenditure ($)	Foot Traffic (number of customers)	Monthly Revenue ($)
Jan	2000	450	15000
Feb	2200	480	15500
Mar	2400	520	16200
Apr	1800	410	14500
May	2500	540	16800
Jun	2300	500	15800
Jul	2100	470	15200
Aug	1900	430	14700
Sep	2600	560	17500
Oct	2500	550	17200
Nov	2300	510	16000
Dec	2100	470	15300

Q.4. Attempt all parts of the following: ***(2×8=16)***

Case Questions should be performed in Excel.

4(a) Compare the impact of advertising expenditure and foot traffic on monthly revenue. Also specify your understanding of correlation and types of correlation.

4(b) following table gives you the regression model

Table- 1: Regression output

Regression Statistics	
Multiple R	0.98
R Square	96%
Adjusted R Square	0.95
Standard Error	210.43
Observations	12

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	9890640.693	4945320.346	111.681	0.000
Residual	9	398525.974	44280.664		
Total	11	10289166.667			

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	5497.403	1275.493	4.310	0.002
Advertising Expenditure ($)	-1.427	2.931	-0.487	0.638
Foot Traffic (number of customers)	27.474	15.569	1.765	0.111

Write the regression equation. If the advertising expenses are $2800 and foot traffic is 602, what would be the Monthly revenue. Also give a detail understanding about regression and its usage in business.